

Xiaoshan Wu

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EDUCATION

Harvard University

Visiting Scholar (Embodied AI & 4D World Models)

– **Advisors:** Yilun Du and Mengyu Wang.

The University of Hong Kong (HKU)

Ph.D. in Electrical and Computer Engineering (Focus: 3D/4D Vision, World Models & Embodied AI)

– **Awards:** Hong Kong PhD Fellowship (HKPFS, Top Tier), HKU Presidential Scholarship.

– **Advisor:** Dr. Xiaojuan Qi. **GPA:** 4.0/4.3.

Zhejiang University (ZJU) & UIUC

Dual B.S. in Computer Engineering (ZJUI Institute)

– **Honors:** Outstanding Graduate (Rank 5), Dean's List. **GPA:** 3.99/4.0 (ZJU), 3.75/4.0 (UIUC).

Cambridge, MA, USA

Apr. 2026 – Aug. 2026

Hong Kong

Sep. 2023 – Present

China & USA

Sep. 2019 – Jun. 2023

RESEARCH INTERESTS

Self-Improving Physical Agents through Interactive 4D World Models. I study how robots can turn purposeful interaction into cumulative, transferable capability. My research combines robust perception, action-responsive 4D foundation world models, and persistent world and experience memory, enabling agents to explore informative interventions in learned surrogate environments, revise strategies from feedback, and transfer the resulting improvements to the physical world under limited interaction budgets.

PUBLICATIONS & PREPRINTS

* indicates equal contribution; † denotes project lead.

1. **Xiaoshan Wu***, R. Cai*, Y. Bao, Z. Su, A. Wu, F. Wu, Z. Li, Z. Qi, H. Chen, W. Ye, J. Zhang, K. Zhou, M. Wang, and Y. Du. **Verifiable 4D World–Action Model for Robust Robot Control.**
Under review at the International Conference on Learning Representations (ICLR), 2027.
2. **Xiaoshan Wu**, X. Lyu, Y. Yu, M. Liu, S. Shi, Z. Wang, and X. Qi. **GaMe: Geometry as Memory for Multi-View Spatial Reasoning.**
Under review at the IEEE International Conference on Robotics and Automation (ICRA), 2027.
3. **Xiaoshan Wu**, X. Lyu, Y. Yu, B. Wang, Z. Wang, and X. Qi. **LiFR-Seg: Anytime High-Frame-Rate Segmentation via Event-Guided Propagation.**
International Conference on Learning Representations (ICLR), 2026.
4. **Xiaoshan Wu**, Y. Yu, X. Lyu, et al. **EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes.**
NeurIPS, 2025 (Spotlight).
5. **Xiaoshan Wu**, W. He, M. Yao, et al. **Event-Based Depth Prediction With Deep Spiking Neural Network.**
IEEE Transactions on Cognitive and Developmental Systems, vol. 16, no. 6, pp. 2008–2018, 2024.
6. Y. Yu*, **Xiaoshan Wu***, X. Hu, et al. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory.**
arXiv:2512.04519, 2025.
7. Y. Yu*, **Xiaoshan Wu*†**, X. Lyu, Z. Wang, and X. Qi. **EVSplat: Uncertainty-Aware Gaussian Splatting for Dynamic Scenes with Event Camera.**
Under review at the AAAI Conference on Artificial Intelligence, 2027.
8. T. Wan*, **Xiaoshan Wu*†**, Y. Yu, et al. **LiFR v2: Completion-Augmented Event Propagation for High-Rate Dense Prediction.**
arXiv:2609.25803, 2026. Under review at IEEE TPAMI.
9. Y. Yu, X. Zhang, S. Wang, ..., **Xiaoshan Wu**, et al. **Efficient and Accurate Neural-Field Reconstruction Using Resistive Memory.**
Nature, vol. 654, pp. 642–651, 2026.
10. M. Liu, X. Lyu, T. Ren, P. Dai, **Xiaoshan Wu**, et al. **FoundationGeo: Learning Spatial Pixel-Wise Fields for Monocular Metric Geometry.**
European Conference on Computer Vision (ECCV), 2026.
11. M. Liu, T. Liu, J. Xia, X. Lyu, **Xiaoshan Wu**, et al. **OptiGeo: Efficient Monocular Geometry for Embodied Perception in Optically Challenging Scenes.**
arXiv:2608.29881, 2026.
12. X. Lyu, M. Liu, **Xiaoshan Wu**, et al. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization.**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7577–7587, 2026.
13. B. Wang, X. Zhang, S. Wang, ..., **Xiaoshan Wu**, et al. **Topology Optimization of Random Memristors for Input-Aware Dynamic SNN.**
Science Advances, vol. 11, no. 16, eads5340, 2025.
14. H. Chen, J. Yang, J. Chen, ..., **Xiaoshan Wu**, et al. **Continuous-Time Digital Twin with Analog Memristive Neural Ordinary Differential Equation Solver.**
Science Advances, vol. 11, no. 22, eadr7571, 2025.
15. C. Li, X. Chen, Z. Zhang, **Xiaoshan Wu**, et al. **Charge-Selective 2D Heterointerface-Driven Multifunctional Floating Gate Memory for In Situ Sensing-Memory-Computing.**
Nano Letters, vol. 24, no. 47, pp. 15025–15034, 2024.

TECHNICAL SKILLS

Research Expertise: Self-Improving Physical Agents, World–Action Models, 3D/4D Vision, Spatial Reasoning, Video Generation, Event Vision.

Frameworks & Tools: PyTorch, CUDA, Diffusers, Open3D, OpenCV.

SELECTED AWARDS

Hong Kong PhD Fellowship (HKPFS)
HKU Presidential Scholarship
Outstanding Graduate of Zhejiang University
NeurIPS Travel Award
ICLR Travel Award

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